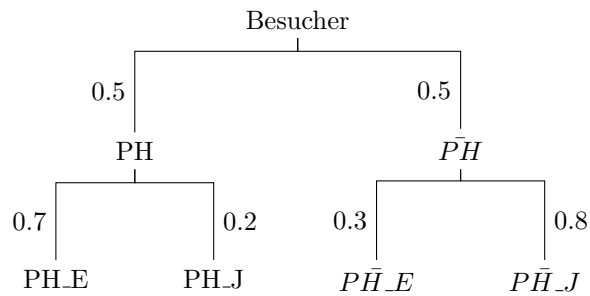


Übungsblatt 3

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December 4, 2025

Teilaufgabe 3.7.2



a)

$$0.5 = 0.7 - 0.7P(J) + 0.2P(J) \leftrightarrow 0.5 = 0.7 - 0.5P(J) \leftrightarrow -0.2 = -0.5P(J) \leftrightarrow 0.4 = P(J) \Rightarrow 40\% \text{ der Besucher sind Jugendliche.}$$

b)

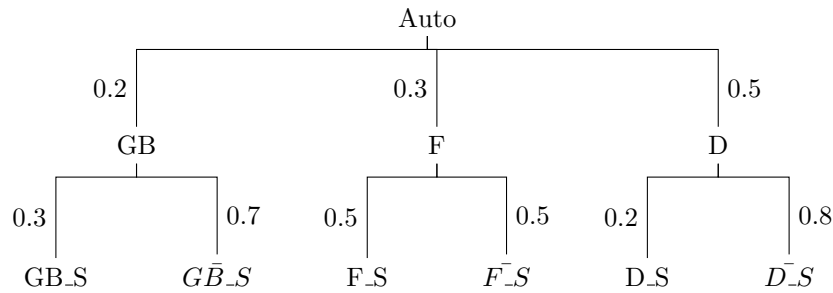
$$P(J|PH) = \frac{0.4 \cdot 0.2}{0.5} = 0.16$$

c)

$$P(E|PH) = \frac{0.6 \cdot 0.7}{0.5} = 0.84$$

Aufgabe 3.9

a)



b)

$$P(S) = P((GB|GB.S) \cup (F|F.S) \cup (D|D.S))$$

$$\rightarrow P(S) = 0.2 \cdot 0.3 + 0.3 \cdot 0.5 + 0.5 \cdot 0.2 = 0.31 = 31\%$$

c)

$$P(D|D.S) = \frac{P(D \cap D.S)}{P(S)} = \frac{0.5 \cdot 0.2}{0.31} \approx 0.323$$

d)

$$P(GB|GB.S) = \frac{P(GB \cap GB.S)}{P(S)} = \frac{0.2 \cdot 0.3}{0.31} = \frac{0.06}{0.31} \approx 0.194$$